1. Alert Priority Levels:
   a) Priority Definitions - Alert priority levels are used to classify security alerts based on their severity and urgency. These levels help Security Operations Centre (SOC) analysts decide which incidents require immediate attention.
      - **Critical:** Indicates severe impact such as data breach, ransomware, or full system compromise. Requires immediate action.
      - **High:** Represents serious threats like unauthorized administrative access or active attack attempts. Needs urgent investigation.
      - **Medium:** Indicates suspicious activity such as multiple failed login attempts. Requires monitoring.
      - **Low:** Refers to low-risk or informational events such as port scanning.

   b) Assignment Criteria-
      Alert prioritization is based on multiple factors-
      - **Asset Criticality:** Systems that are essential to operations are prioritized over non-critical systems.
      - **Exploit Likelihood:** Vulnerabilities with publicly available exploits are considered higher risk.
      - **Business Impact:** Incidents that can cause financial loss, reputational damage, or service disruption are assigned higher priority.

   c) Scoring Systems-

The **Common Vulnerability Scoring System (CVSS)** is used to measure and quantify the severity of vulnerabilities.

- 9.0 – 10.0 → Critical
- 7.0 – 8.9 → High
- 4.0 – 6.9 → Medium
- 0 – 3.9 → Low

CVSS is based on three types of metrics:

- **Base Metrics:** Inherent characteristics of the vulnerability
- **Temporal Metrics:** Current exploit activity and availability
- **Environmental Metrics:** Impact on a specific organization

2. Incident Classification:

a) Incident Categories-

Incidents are classified based on their type:

- **Malware:** Virus, trojan, ransomware
- **Phishing:** Fraudulent emails or links
- **DDoS:** Traffic flooding attacks
- **Insider Threat:** Unauthorized actions by employees
- **Data Exfiltration:** Unauthorized data transfer

b) Taxonomy-

Standard frameworks are used for classification:

- MITRE ATT&CK → Maps attacks to techniques
- ENISA Incident → Standard incident categories
- VERIS → Structured incident reporting

c) Contextual Metadata-

Incidents are enriched with additional details:

- Affected systems
- Timestamp
- Source IP address
- Indicators of Compromise (IOCs) such as file hash

3. Basic Incident Response:

a) Incident Lifecycle-

Incident response follows a structured lifecycle:

- **Preparation:** Establish tools, policies, and playbooks
- **Identification:** Detect and analyse security incidents
- **Containment:** Isolate affected systems to prevent spread
- **Eradication:** Remove malware or eliminate the threat
- **Recovery:** Restore systems and resume normal operations
- **Lessons Learned:** Review the incident and improve processes

b) Procedures-

Following procedures followed during incident response:

- **System Isolation:** Disconnect compromised systems
- **Evidence Preservation:** Collect memory dumps and calculate file hashes
- **Communication Protocols:** Inform relevant teams and stakeholders
- **Automation Tools:** Use tools like Splunk Phantom for automated workflows

# ALERT MANAGEMENT PRACTICE

1. Alert Classification System:
   - An alert classification system was created using Google Sheets to categorize security alerts based on type, priority, CVSS score, and mapped MITRE ATT&CK techniques. This helps in organizing alerts and understanding their severity.
   - The classification includes different types of attacks such as phishing, exploitation, port scanning, and brute-force attempts. Each alert is assigned a priority level based on its impact and CVSS score.
   - Example-
     - Phishing Email → High Priority → T1566

J14

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | ALERT ID | TYPE | PRIORITY | CVSS | MITRE | |
| 2 | 001 | Phishing Email | High | 7.5 | T1566 | |
| 3 | 002 | Log4Shell Exploit | Critical | 9.8 | T1190 | |
| 4 | 003 | Port Scan | Low | 3.1 | T1046 | |
| 5 | 004 | SSH Brute Force | Medium | 6.5 | T1110 | |
| 6 | 005 | Phishing Suspicious Link | High | 7.8 | T1566 | |
| 7 | | | | | | |

Alert Classification Table with Priority and MITRE Mapping

This structured classification improves visibility of threats and enables faster decision-making by SOC analysts during incident handling.

2. Prioritize Alerts using CVSS:

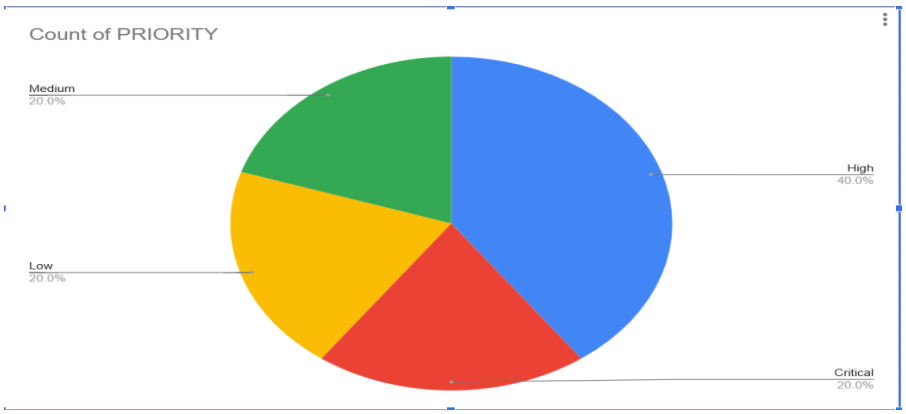
   Alerts were prioritized using CVSS scores to determine their severity:

   - Critical alerts (9.0–10.0) require immediate action
   - High alerts (7.0–8.9) require urgent attention
   - Medium alerts (4.0–6.9) require monitoring
   - Low alerts (0–3.9) are informational

For example, the Log4Shell has a CVSS score of 9.8, making it a critical alert.

3. Dashboard Creation:

   - A dashboard was created using Google Sheets to visualize alert priorities. A pie chart was used to represent the distribution of alerts across different priority levels such as Critical, High, Medium, and Low.
   - This visualization helps SOC analysts quickly understand the proportion of high-risk alerts and prioritize response actions accordingly.




Alert Priority Distribution Dashboard

4. Incident Ticket (TheHive):
    - An incident ticket was created to document a critical security event using structured fields such as title, description, priority, and assignee. This helps in tracking the incident lifecycle from detection to resolution.
    - The ticket includes important details like indicators of compromise (IOCs), affected systems, and severity level. It ensures proper assignment to SOC analysts and enables efficient incident handling and escalation.
    - This process improves coordination among security teams and ensures timely response to high-priority incidents.

5. Escalation Role-Play:

    - An escalation process was performed to notify higher-level (Tier 2) security analysts about a critical incident. The escalation includes key details such as incident severity, indicators of compromise (IOCs), and potential impact.
    - This ensures that critical threats are handled by experienced personnel and appropriate actions are taken quickly to minimize damage.
    - Effective escalation improves response time and ensures proper coordination between different SOC levels.

# RESPONSE DOCUMENTATION

1. Incident Response Template:

- Executive Summary-

  A phishing incident was simulated where a user received a malicious email containing a suspicious link. The user interaction triggered an alert in the SIEM system. The incident was investigated and appropriate response actions were taken to mitigate the threat.

- Timeline-

| TIMESTAMP | EVENT |
|---|---|
| 2025-08-18 14:00:00 | Phishing Email received |
| 2025-08-18 14:05:00 | User clicked malicious link |
| 2025-08-18 14:10:00 | Alert generated (SIEM) |
| 2025-08-18 14:15:00 | Investigation started |
| 2025-08-18 14:30:00 | Endpoint isolated |

- Impact Analysis-

  The phishing attack had the potential to compromise user credentials and system security. If not detected early, it could lead to unauthorized access and data leakage.

- Remediation Steps-
  - Isolated the affected endpoints
  - Blocked malicious URL
  - Monitored system for further activity

- Lessons Learned-

  The incident highlighted the importance of user awareness and timely detection. Strengthening email filtering and security monitoring can help prevent such attacks in the future.

2. Investigation Steps:

| TIMESTAMP | ACTION |
|---|---|
| 2025-08-18 14:00:00 | Isolated endpoint |
| 2025-08-18 14:30:00 | Collected memory dump |

These steps document the actions taken during the investigation process.

3. Phishing Checklist:
   - ✓ Confirm email headers
   - ✓ Check link reputation
   - ✓ Identify affected users

This checklist ensures all necessary validation steps are followed during phishing analysis.

4. Post-Mortem:

   The incident revealed gaps in user awareness and email security controls. Early detection helped prevent serious impact. Improvements such as stronger email filtering, user training, and faster response mechanisms are required. Regular monitoring and incident response drills can further enhance security and reduce the risk of similar attacks.

# ALERT TRIAGE PRACTICE

1. Triage Simulation:

   A mock alert related to brute-force SSH attempts was analysed using Wazuh. The alert indicated multiple failed login attempts from a suspicious IP address, suggesting a potential brute-force attack.

| Alert ID | Description | Source IP | Priority | Status |
|---|---|---|---|---|
| 002 | Brute-force SSH Attack | 192.168.1.100 | Medium | Open |

Brute-force SSH Alert Details

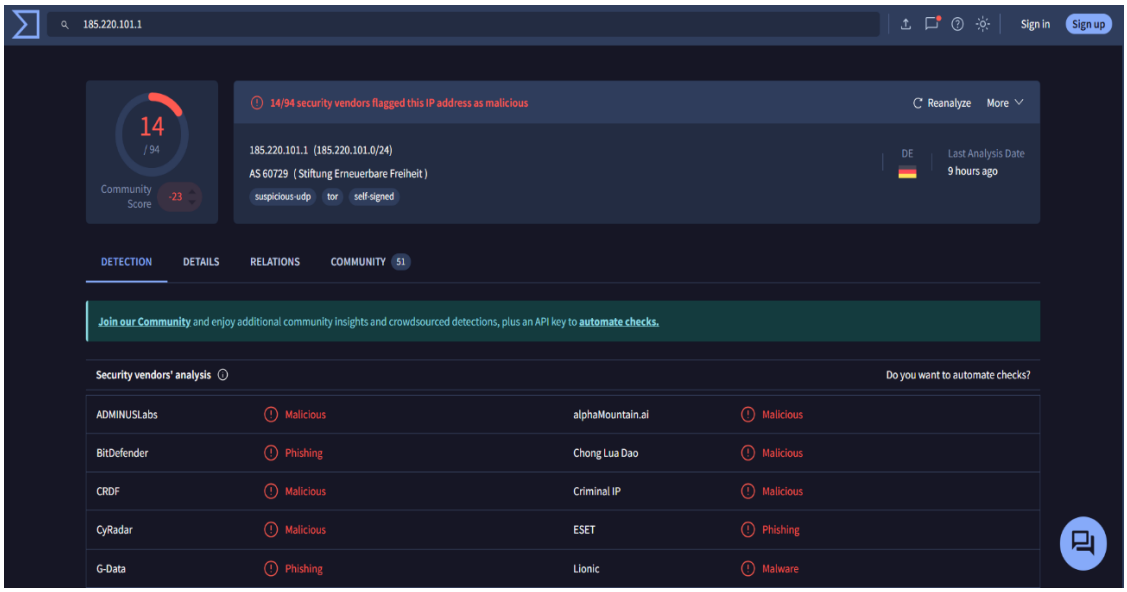
2. Analysis of Alert:

   The alert logs showed multiple failed login attempts using different usernames such as admin, root, and test from the same IP address. This pattern confirms a brute-force attempt targeting SSH services.

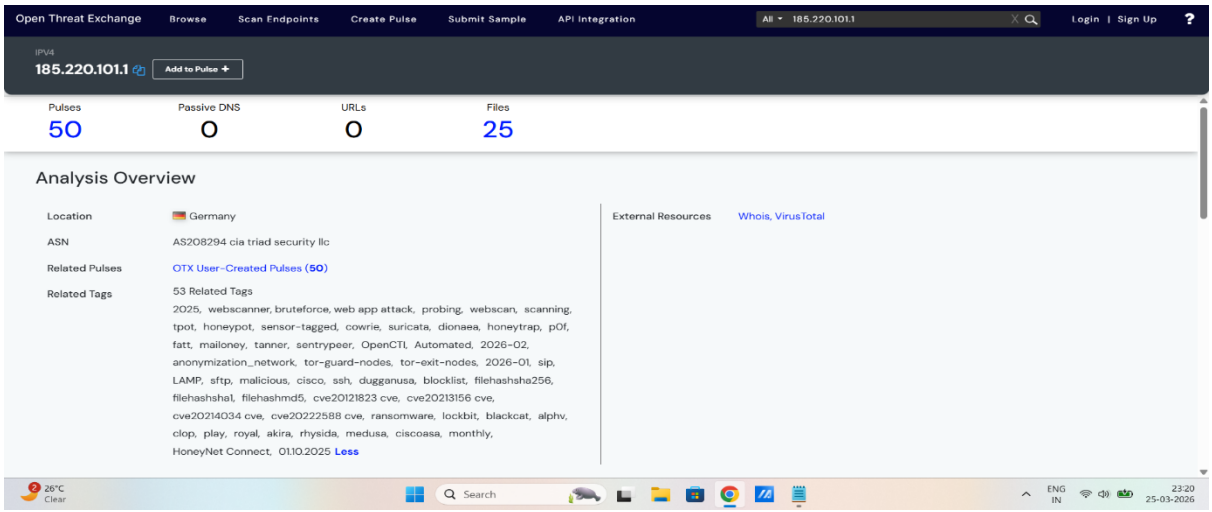
```
Mar 25 10:15:23 sshd[1024]: Failed password for invalid user admin from 185.220.101.1 port 22
Mar 25 10:15:25 sshd[1024]: Failed password for invalid user root from 185.220.101.1 port 22
Mar 25 10:15:30 sshd[1024]: Failed password for invalid user test from 185.220.101.1 port 22
Mar 25 10:15:35 sshd[1024]: Failed password for invalid user guest from 185.220.101.1 port 22
Mar 25 10:15:40 sshd[1024]: Connection closed by 185.220.101.1 port 22 [preauth]
```

Multiple Failed SSH Login Attempts Indicating Brute-force Attack

3. Threat Intelligence Validation:
   The IP address was analysed using threat intelligence platforms and was flagged as malicious by multiple security vendors. It is associated with brute-force and scanning activities. This confirms that the alert is a true positive and the source IP is potentially dangerous. Used both: **192.168.1.100 and 185.220.101.1**




IP Reputation Analysis using VirusTotal




Threat Intelligence from AlienVaultOTX

# EVIDENCE PRESERVATION

1.Volatile Data Collection:

Volatile data was collected from the system using tools like Velociraptor. Network connection details were extracted using netstat commands to identify active and suspicious connections. This data is important as it exists only in memory and can be lost after system shutdown.

```
Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       1800
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:3306           0.0.0.0:0              LISTENING       6032
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       8560
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       15556
  TCP    0.0.0.0:33060          0.0.0.0:0              LISTENING       6032
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       1480
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       1308
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING       2264
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING       3480
```

All Listening Ports

```
  TCP    127.0.0.1:6789         127.0.0.1:6790         ESTABLISHED     20352
  TCP    127.0.0.1:6790         127.0.0.1:6789         ESTABLISHED     20352
  TCP    127.0.0.1:6793         127.0.0.1:6794         ESTABLISHED     20352
  TCP    127.0.0.1:6794         127.0.0.1:6793         ESTABLISHED     20352
  TCP    127.0.0.1:6800         0.0.0.0:0              LISTENING       3388
  TCP    127.0.0.1:6800         127.0.0.1:13441        ESTABLISHED     3388
  TCP    127.0.0.1:13441        127.0.0.1:6800         ESTABLISHED     10140
```

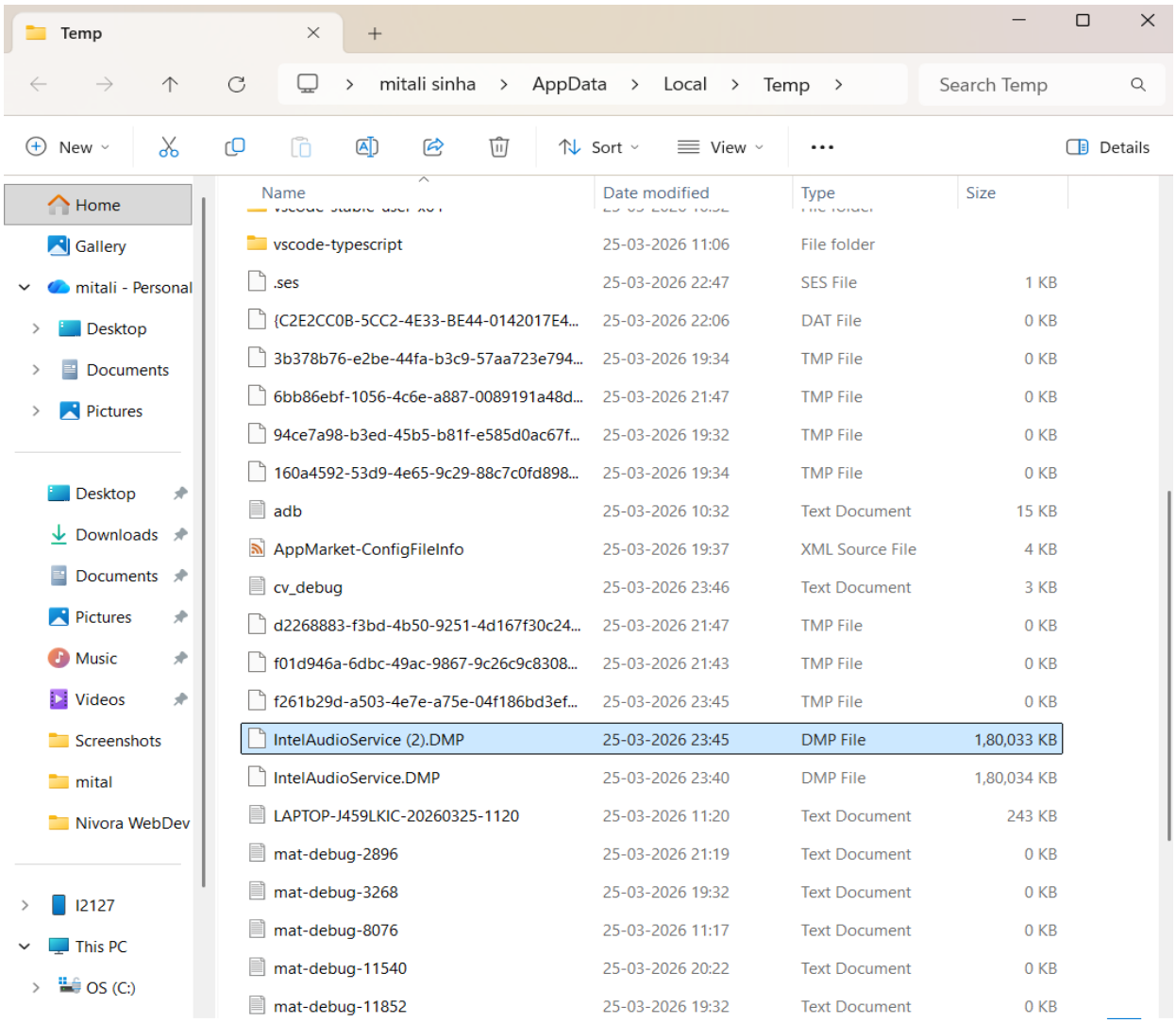
Localhost Connection

```
  TCP    192.168.1.13:3360      172.64.146.169:443     ESTABLISHED     4544
  TCP    192.168.1.13:4683      172.253.118.188:443    ESTABLISHED     4544
  TCP    192.168.1.13:6814      48.218.107.32:443      ESTABLISHED     2448
  TCP    192.168.1.13:13492     172.64.148.235:443     ESTABLISHED     4544
  TCP    192.168.1.13:14684     142.250.194.228:443    TIME_WAIT       0
  TCP    192.168.1.13:15257     162.159.152.4:443      ESTABLISHED     4544
  TCP    192.168.1.13:17692     52.123.253.89:443      ESTABLISHED     5068
  TCP    192.168.1.13:27810     40.104.77.130:443      TIME_WAIT       0
  TCP    192.168.1.13:27813     104.18.21.213:80       TIME_WAIT       0
  TCP    192.168.1.13:35496     20.192.44.82:443       ESTABLISHED     13608
  TCP    192.168.1.13:49422     4.213.25.240:443       ESTABLISHED     5256
  TCP    192.168.1.13:55239     48.218.104.163:443     ESTABLISHED     3268
  TCP    192.168.1.13:57480     150.109.28.54:443      ESTABLISHED     10140
```

External IP Connections

2. Evidence Collection:

A memory dump of the system was collected to preserve evidence of the incident. Memory acquisition helps in analysing running processes, malicious code, and system activity at the time of the attack.




Memory Dump File Collected from System

3.Hash Calculation:

The integrity of the collected evidence was ensured by generating a SHA256 hash. Hashing helps verify that the evidence has not been altered during analysis.

```
SHA256 hash of C:\Users\mital\AppData\Local\Temp\IntelAudioService (2).DMP:
dcd6ec97da4aa73cbd19b3895d5927d16f2e50d4e0720d946c8ed8e6ef24adf0
```

SHA256 Hash of Memory Dump File

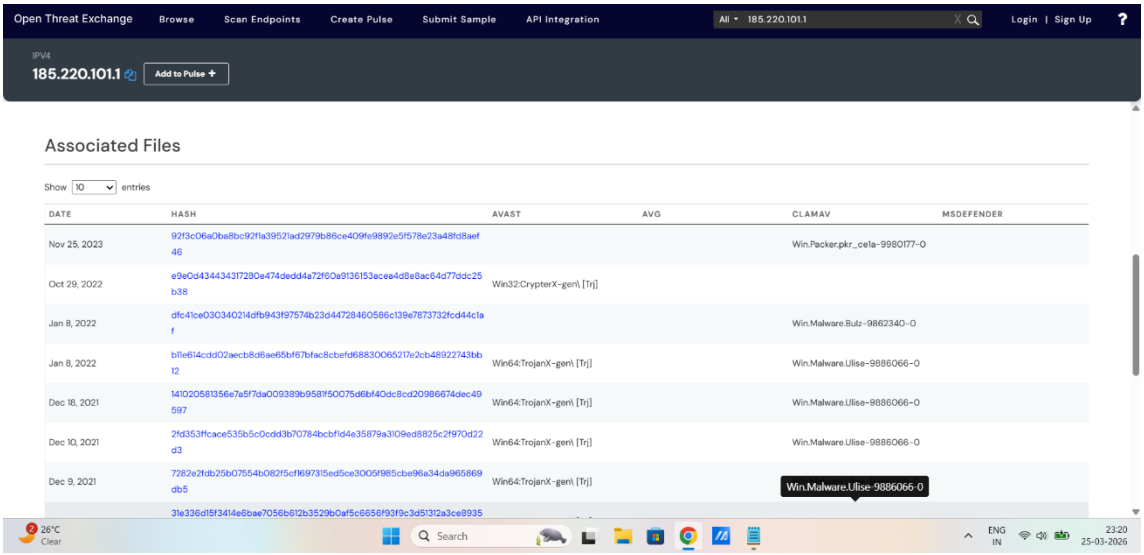
4. Evidence Documentation:

| | |
|---|---|
| ITEM | Memory Dump |
| DESCRIPTION | Server-X Dump |
| BY | SOC Analyst |
| DATE | 2025-08-18 |
| HASH VALUE | dcd6ec97da4aa73cbd19b3895d5927d16f2e50d4e0720d946c8ed8e6ef24adf0 \| |

Proper documentation ensures chain of custody and maintains the integrity of digital evidence.

5. Forensic Analysis:

The collected evidence was further analysed using threat intelligence tools. File hashes were checked against multiple security vendors, and several detections indicated the presence of malicious files associated with the incident.




Malware Detection using Hash-based Analysis

# CAPSTONE PROJECT

1.Attack Simulation:

A simulated attack was performed using Metasploit by exploiting a vulnerable VSFTPD service. The exploit vsftpd_234_backdoor was used to gain unauthorized access to the target system, demonstrating an initial access attack scenario.

2. Detection and Triage:

| Timestamp | Source IP | Alert Description | MITRE Technique |
|---|---|---|---|
| 2025-08-18 11:00:00 | 192.168.1.100 | VSFTPD exploit | T1190 |

The attack was detected using Wazuh, which generated alerts indicating suspicious activity related to the exploited service.

3.Response:

The affected system was isolated to prevent further compromise. The attacker's IP address was blocked using CrowdSec, and connectivity was tested using ping to ensure the block was successful.

4.Report:

- A simulated cyberattack was conducted to evaluate the system's security posture. The attack targeted a vulnerable VSFTPD service using Metasploit and successfully demonstrated unauthorized access. The activity was detected by the monitoring system, which generated alerts indicating suspicious behaviour.
- The incident was analysed, and it was confirmed that the attack followed an initial access technique mapped to MITRE ATT&CK T1190. Immediate response actions were taken, including isolating the affected system and blocking the attacker's IP address using CrowdSec.
- No real damage occurred as this was a controlled simulation. However, the test highlighted the importance of continuous monitoring, timely detection, and effective response mechanisms. It also emphasized the need for regular patching of vulnerable services.
- Recommendations include strengthening intrusion detection systems, applying security updates regularly, and implementing strict access controls to prevent unauthorized access. Overall, the response was effective and demonstrated the organization's capability to handle security incidents efficiently.

5. Stakeholder Briefing:

   - A simulated cyberattack was carried out to test system security. The attack exploited a vulnerability in the FTP service and demonstrated unauthorized access. The monitoring system successfully detected the activity, and immediate actions were taken to isolate the affected system and block the attacker.
   - No actual data was compromised as this was a controlled test. The incident highlights the importance of regular system updates and strong monitoring systems. Overall, the response was effective, and necessary steps have been identified to further strengthen system security and prevent similar incidents in the future.

# CONCLUSION

- This project demonstrated a complete SOC workflow, including alert management, incident classification, response documentation, alert triage, and evidence preservation. Alerts were effectively prioritized using CVSS and mapped to frameworks like MITRE ATT&CK for accurate classification.
- Practical tasks such as analysing brute-force attacks, validating threats using VirusTotal and AlienVault OTX, and preserving digital evidence helped in understanding real-world security operations.
- The capstone project successfully simulated an end-to-end attack lifecycle, from detection to response and reporting.